PIYUSH CHHALLARE

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EDUCATION

Northeastern University, Boston, MA **May 2028**

Master of Science, Semiconductor Engineering, Concentration in Devices and Nano systems

Relevant Coursework: RFIC, Analog IC, MEMS, Computer Architecture

Purdue University, West Lafayette, IN **May 2026**

Bachelor of Science, Physics, Quantum Information Science and Technology (QIST) Certificate

Relevant Coursework: Semiconductor Physics, Solid State Physics, Quantum Physics, Electronics, Optical Physics, Plasma Physics

TECHNICAL SKILLS & CERTIFICATIONS

- **Fabrication Tools & Process**: Photolithography, CVD, ALD, MBE, PECVD, RIE, ICP, Diffusion, Oxidation, Ion Implantation, DOE, SPC
- **Characterization**: SEM, TEM, AFM, XRD, XRF, Hall Measurements, MPMS, PPMS
- **Hardware Engineering**: Oscilloscope, Signal Integrity Debugging, Logic Analyzer, Test Benching, VLSI, FPGA, ASIC
- **Software**: Python, C++, R, Cadence, ADS, Electric EDA, Monte Carlo, Data Analysis, Algorithm Development, Statistical Modeling, Jupyter Notebook, Origin, MS-Office
- **Certifications**: Fundamentals of VLSI Design, Semiconductor Characterization: MOSFETs, Diodes, Optical & X-ray, Electron & Ion beam, Semiconductor Characterization 101

PROFESSIONAL EXPERIENCE

Northeastern University, Boston, MA **Sep 2026 – Present**

Graduate Student Researcher, Ebrahimi Lab

- Developing an adaptive RFIC-based control framework for superconducting transmon qubits to maintain high gate fidelity under control errors and device variations.
- Modeling multi-level transmon dynamics in Python/QuTiP to quantify degradation of gate fidelity due to error mechanisms.
- Simulating DRAG-shaped microwave pulses and closed-loop calibration schemes to correct qubit-control errors.
- Evaluating RFIC and cryo-CMOS architectures for implementing quantum control algorithms in scalable cryogenic hardware.

Purdue University, West Lafayette, IN **May 2024 – Sep 2026**

Undergraduate Physics Researcher, Banerjee Lab

- Synthesized 20+ quantum materials via flux growth and CVD and used XRD to verify over 90% disorder-free grown samples.
- Measured magnetic and transport properties of grown samples down to sub-Kelvin temperatures using MPMS and PPMS to map field-dependent quantum phases in low-dimensional systems.
- Fabricated low-temperature measurement devices using photolithography, ALD and RIE for low-temperature Hall studies.
- Automated measurement data processing by building a Python analysis pipeline, reducing analysis time by ~30% across multiple datasets.
- Co-authored a research paper on the triangular-lattice quantum spin liquid candidate $TlYbS_2$ (arXiv:2609.11881, 2026), contributing toward crystal growth and low-temperature magnetic measurements.

The Data Mine, West Lafayette, IN **Aug 2023 – Dec 2023**

Undergraduate Researcher

- Modeled top quark decay processes from CMS open data using Python-based Monte Carlo Simulations and compared with experimental distributions.
- Built Python event-selection pipelines to separate W and Z boson decays across recorded collision events for analysis of particle decay channels.
- Diagnosed anomalies in experimental data through statistical checks, tracing their sources to improve the reliability of downstream analysis.

PROJECTS

Magnetic Properties of Novel Triangular Quantum Magnets **Aug 2024 – Sep 2026**

Banerjee Lab, Purdue University

- Synthesized ytterbium-based delafossites via solid-state reaction, flux growth, and verified phase purity of samples for low-temperature study.
- Measured magnetic properties using MPMS and PPMS to identify quantum spin liquid candidates with no long-range ordering down to 1.8K.

PUBLICATIONS

- B. P. Belbase, A. Unnikrishnan, **P. Chhallare**, M. B. Neupane, E. S. Choi, S. Erdmann, M. U. Akbar, M. Hemmida, H.-A. Krug von Nidda, P. Gegenwart, and A. Banerjee, "Low temperature thermodynamics of S_{eff} = 1/2 triangular lattice quantum spin liquid candidate $TlYbS_2$," arXiv:2609.11881 (2026).